

GBU 2301: Business Statistics

Week 4: Introduction to Probability

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Required reading: Anderson et al., Statistics for Business and Economics, 15th ed., Chapter 4

Outline

- Random Experiments and Counting Rules
- Assigning Probabilities and Events
- Basic Relationships of Probability
- Conditional Probability and Independence
- Bayes' Theorem

Random Experiments and Sample Spaces

- A random experiment produces one uncertain outcome. The sample space $\mathcal{S}$ contains all possible outcomes; an event is a subset of $\mathcal{S}$.
- For an order, record whether delivery is late (L) and whether a complaint occurs (C).

$$\mathcal{S} = \{(L, C), (L, C^c), (L^c, C), (L^c, C^c)\}$$

- A^c is the event that A does not occur.
- The four pairs are possible outcomes.
- The pairs need not have equal probabilities.

Counting Rule for Multiple Steps

For k steps with $n_1, n_2, \dots, n_k$ choices, the number of outcomes is $n_1 n_2 \cdots n_k$.

- A sales offer has three channels, two delivery options, and two payment methods.

$$3 \times 2 \times 2 = 12 \text{ possible combinations}$$

- Choose one option at each step.
- Multiply the number of choices available at each step.

Permutations and Combinations

Select r objects from n distinct objects, where $0 \leq r \leq n$.

Counting rule	Does order matter?	Number of selections
Permutation	Yes	$P(n, r) = \frac{n!}{(n-r)!}$
Combination	No	$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Each unordered selection has $r!$ possible orders: $P(n, r) = r!C(n, r)$.

Permutations and Combinations

- A firm has six candidates.
- Appoint a lead and a backup. The roles make order matter:

$$P(6, 2) = \frac{6!}{(6-2)!} = 6(5) = 30$$

- Select a three-person committee. Rearranging the same three people does not form a new committee:

$$C(6, 3) = \binom{6}{3} = \frac{6!}{3!(6-3)!} = 20$$

Exercise 1

A firm has six candidates.

- How many three-person shortlists are possible?
- How many contain one specified candidate?
- Separately, how many lead-and-backup assignments are possible from all six?

Exercise 1: Solution

$$\text{All shortlists} = \binom{6}{3} = 20$$

Fix the specified candidate and choose two of the other five:

$$\text{Shortlists containing that candidate} = \binom{5}{2} = 10$$

Lead and backup are distinct roles:

$$\text{Assignments} = P(6, 2) = 6(5) = 30$$

Assigning Probabilities

Basis	Numerical example
Classical, equally likely outcomes	One face of a fair die: $1/6$
Relative frequency	45 complaints in 200 orders: $45/200 = 0.225$
Subjective assessment	An analyst assigns a 0.30 chance to approval

- Every assigned probability lies between 0 and 1.
- Probabilities across the full sample space sum to 1.

Probability of an Event

For sample points ω_i in $\mathcal{S}$ and an event $A \subseteq \mathcal{S}$:

$$P(\omega_i) \geq 0, \quad \sum_{\omega_i \in \mathcal{S}} P(\omega_i) = 1, \quad P(A) = \sum_{\omega_i \in A} P(\omega_i)$$

If the N sample points are equally likely, $P(A) = |A|/N$. From n observed trials, the relative-frequency estimate is n_A/n .

Orders: Counts

A retailer records 200 orders.

Delivery	Complaint	No complaint	Total
Late	30	20	50
On time	15	135	150
Total	45	155	200

- Let L denote a late order and C a complaint.
- For row i and column j , with cell count n_{ij} and total n :

$$p_{ij} = \frac{n_{ij}}{n}, \quad p_{i\cdot} = \sum_j p_{ij}, \quad p_{\cdot j} = \sum_i p_{ij}$$

Delivery	C	C^c	Total
L	0.150	0.100	0.250
L^c	0.075	0.675	0.750
Total	0.225	0.775	1.000

- The interior cells are joint probabilities.
- The row and column totals are marginal probabilities.

Events and Their Probabilities

From the order table:

$$P(L) = \frac{50}{200} = 0.25, \quad P(C) = \frac{45}{200} = 0.225$$

Both events occur for 30 orders:

$$P(L \cap C) = \frac{30}{200} = 0.15$$

Event operations select cells of the crosstabulation:

Operation	Meaning	Selected order counts
$A \cap B$	Both events	$L \cap C : 30$
$A \cup B$	At least one event	$L \cup C : 30 + 20 + 15 = 65$
A^c	Event A does not occur	$L^c : 15 + 135 = 150$

Exercise 2

A bank reviewed 240 loan applications.

Application channel	Approved	Declined	Total
Online	24	76	100
Branch	84	56	140
Total	108	132	240

An analyst uses $24/100$ to report that 24% of all applications were both online and approved.

- Is this correct?
- Calculate the share of all applications that were both online and approved and the overall approval share. What does $24/100$ measure?

Exercise 2: Solution

$$P(\text{online} \cap \text{approved}) = \frac{24}{240} = 0.10$$

$$P(\text{approved}) = \frac{24 + 84}{240} = \frac{108}{240} = 0.45$$

The claim is incorrect. $24/100 = 0.24$ is the approval share among online applications, not the share of all applications that were both online and approved.

Complement Rule

- An event and its complement cover the full sample space without overlap.

$$P(A^c) = 1 - P(A)$$

- For the order table:

$$P(L^c) = 1 - 0.25 = 0.75, \quad P(C^c) = 1 - 0.225 = 0.775$$

Addition Rule

- For any two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Orders that were late or generated a complaint include 30 orders counted in both groups.

$$\begin{aligned} P(L \cup C) &= P(L) + P(C) - P(L \cap C) \\ &= 0.25 + 0.225 - 0.15 = 0.325 \end{aligned}$$

- Check by counting unique orders: $(50 + 45 - 30)/200 = 65/200 = 0.325$.

Mutually Exclusive Events

- Events A and B are mutually exclusive when $A \cap B = \emptyset$. Then,

$$P(A \cup B) = P(A) + P(B)$$

- Late and on-time delivery cannot occur for the same order:

$$P(L \cap L^c) = 0, \quad P(L \cup L^c) = P(L) + P(L^c) = 1$$

- Late delivery and a complaint can occur together.
- Their overlap must be subtracted in the addition rule.

Exercise 3

A retailer reviewed 600 shipments.

Delivery	Damaged	Not damaged	Total
Delayed	24	96	120
On time	36	444	480
Total	60	540	600

- The retailer gives one 30-MAD voucher for each shipment that was delayed or damaged.
- A spreadsheet budgets $30(120 + 60)$ MAD.
- Audit the budget. How many shipments had exactly one problem, and what proportion receive no voucher?

Exercise 3: Solution

$$\text{Shipments with at least one problem} = 120 + 60 - 24 = 156$$

$$\text{Voucher cost} = 156(30) = 4,680 \text{ MAD}$$

The spreadsheet counts the 24 delayed and damaged shipments twice, overstating the budget by $24(30) = 720$ MAD.

$$\text{Exactly one problem} = 96 + 36 = 132$$

$$P(\text{no voucher}) = \frac{444}{600} = 0.74$$

Conditional Probability

For $P(B) > 0$, $P(A | B) = \frac{P(A \cap B)}{P(B)}$.

The condition changes the denominator to the group known to have occurred.

Among the 50 late orders, 30 generated complaints. Among the 150 on-time orders, 15 generated complaints.

$$P(C | L) = \frac{P(L \cap C)}{P(L)} = \frac{30}{50} = 0.60$$

$$P(C | L^c) = \frac{P(L^c \cap C)}{P(L^c)} = \frac{15}{150} = 0.10$$

The full-sample calculation $30/200 = 0.15$ is $P(L \cap C)$: the probability that an order was both late and associated with a complaint.

Reversing the Condition

For events A and B with positive probabilities:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

For the orders:

$$P(L \mid C) = \frac{0.15}{0.225} \approx 0.667, \quad P(C \mid L) = \frac{0.15}{0.25} = 0.60$$

$P(L \mid C)$ describes late deliveries among complaints; $P(C \mid L)$ describes complaints among late deliveries.

Multiplication Rule

From the definition of conditional probability, for $P(A) > 0$:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Multiply both sides by $P(A)$:

$$P(A)P(B \mid A) = P(A \cap B)$$

For the order table:

$$P(L \cap C) = P(L)P(C \mid L) = 0.25(0.60) = 0.15$$

Multiplication Rule: Example

A batch has 10 invoices: 3 overdue and 7 current. Draw two uniformly at random, one after the other, without replacement.

- What is the probability that both are overdue?

Let A denote an overdue first selection and B an overdue second selection.

After an overdue invoice is selected first, 2 of the remaining 9 are overdue.

$$P(A) = \frac{3}{10}, \quad P(B | A) = \frac{2}{9}$$

$$P(A \cap B) = P(A)P(B | A) = \frac{3}{10} \frac{2}{9} = \frac{1}{15} \approx 0.0667$$

- Check by counting first-second draw sequences.
 - There are $10(9) = 90$ possible sequences, and $3(2) = 6$ have both invoices overdue.

$$P(\text{both overdue}) = \frac{6}{90} = \frac{1}{15}$$

Partition of the Sample Space

A partition separates all outcomes into nonoverlapping groups.

- For events $A_1, \dots, A_k$ with $P(A_i) > 0$:

$$A_i \cap A_j = \emptyset \quad (i \neq j), \quad A_1 \cup \dots \cup A_k = S$$

- Any event B can be split into disjoint parts: $B = \bigcup_{i=1}^k (B \cap A_i)$.

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $B = \{2, 4, 6, 8, 10\}$.

Partition group	Elements	Elements also in B
A_1	$\{1, 2, 3, 4\}$	$\{2, 4\}$
A_2	$\{5, 6, 7, 8, 9, 10\}$	$\{6, 8, 10\}$

$A_1 \cap A_2 = \emptyset$, $A_1 \cup A_2 = S$, and the two parts of B contain $2 + 3 = 5$ elements.

Late (L) and on-time (L^c) orders form a partition.

- Every order is in exactly one group.
- Therefore every complaint is in exactly one of these two parts:

$$C = (C \cap L) \cup (C \cap L^c)$$

Total Probability

The two parts $C \cap L$ and $C \cap L^c$ cannot occur together, so their probabilities add:

$$P(C) = P(C \cap L) + P(C \cap L^c)$$

Apply the multiplication rule to each part:

$$\begin{aligned} P(C) &= P(L)P(C \mid L) + P(L^c)P(C \mid L^c) \\ &= 0.25(0.60) + 0.75(0.10) = 0.150 + 0.075 = 0.225 \end{aligned}$$

For any event B and partition $A_1, \dots, A_k$, the general result is

$$P(B) = \sum_{i=1}^k P(B \cap A_i) = \sum_{i=1}^k P(A_i)P(B \mid A_i)$$

Independence

Events A and B are independent when $P(A \cap B) = P(A)P(B)$. If $P(B) > 0$, this is equivalent to $P(A | B) = P(A)$.

If L and C were independent, knowing L would not change the probability of C .

$$P(C | L) = 0.60 \neq P(C) = 0.225$$

Equivalently,

$$P(L \cap C) = 0.15 \neq P(L)P(C) = 0.25(0.225) = 0.05625$$

Late delivery and complaints are not independent in these records.

In the ten-element example, assume each element is equally likely. For $A_1 = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8, 10\}$:

$$P(A_1 \cap B) = \frac{2}{10} = 0.20 = P(A_1)P(B) = \frac{4}{10} \frac{5}{10}$$

Thus A_1 and B are independent.

Independence and Mutual Exclusiveness

L and L^c are mutually exclusive:

$$P(L \cap L^c) = 0$$

They are not independent:

$$P(L)P(L^c) = 0.25(0.75) = 0.1875 \neq 0$$

Mutual exclusiveness concerns overlap. Independence concerns whether one event changes the probability of the other.

In the same ten-element example, A_1 and A_2 are mutually exclusive, but

$$P(A_1 \cap A_2) = 0 \neq P(A_1)P(A_2) = \frac{4}{10} \frac{6}{10} = 0.24$$

so they are not independent.

Exercise 4

A campaign report claims that exposure and purchase are independent.

Group	Purchased	Did not purchase	Total
Exposed	48	72	120
Not exposed	16	64	80
Total	64	136	200

Assess the claim numerically. What does the table establish about the effect of the campaign?

Exercise 4: Solution

$$P(\text{purchase} \mid \text{exposed}) = \frac{48}{120} = 0.40$$

$$P(\text{purchase} \mid \text{not exposed}) = \frac{16}{80} = 0.20$$

$$P(\text{purchase}) = \frac{64}{200} = 0.32$$

The rates differ, so purchase and exposure are not independent in these observed data. The table alone does not show that exposure caused the difference.

Bayes' Theorem

Suppose $A_1, \dots, A_k$ partition the sample space, with $P(A_i) > 0$, and event B has occurred, with $P(B) > 0$. We want the probability of group A_i after observing B .

Start with conditional probability:

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}$$

The multiplication rule and total probability give

$$P(A_i \cap B) = P(A_i)P(B | A_i), \quad P(B) = \sum_{j=1}^k P(A_j)P(B | A_j)$$

Substituting both expressions:

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^k P(A_j)P(B | A_j)}$$

- $P(A_i)$ is the prior probability.
- $P(B | A_i)$ is the chance of observing B within that group.
- $P(A_i | B)$ is the updated or posterior probability.

Bayes' Theorem: Orders

Delivery	Prior	Complaint rate	Joint probability
Late	0.25	0.60	$0.25(0.60) = 0.150$
On time	0.75	0.10	$0.75(0.10) = 0.075$
Total	1.00		$P(C) = 0.225$

$$P(L | C) = \frac{0.150}{0.225} \approx 0.667$$

Inspection Example

A screening system flags defective units.

Quantity	Probability
Unit is defective	0.10
Flag given defective	0.80
Flag given acceptable	0.20

What is the probability that a flagged unit is defective?

Inspection Example: Positive Flag

Expected counts for 1,000 units under the given probabilities:

Actual status	Flagged	Not flagged	Total
Defective	80	20	100
Acceptable	180	720	900
Total	260	740	1,000

$$P(\text{defective} \mid \text{flag}) = \frac{80}{260} \approx 0.308$$

The same result follows from $0.10(0.80)/[0.10(0.80) + 0.90(0.20)]$.

Inspection Example: No Flag

A unit without a flag can still be defective.

$$P(\text{defective} \mid \text{no flag}) = \frac{20}{740} \approx 0.027$$

Equivalently,

$$\frac{0.10(0.20)}{0.10(0.20) + 0.90(0.80)} \approx 0.027$$

The Base Rate

Keep the same inspection accuracy, but suppose the defect rate is 25 percent.

$$P(\text{defective} \mid \text{flag}) = \frac{0.25(0.80)}{0.25(0.80) + 0.75(0.20)} = \frac{0.20}{0.35} \approx 0.571$$

- With a 10 percent defect rate, the result was 0.308.
- The test did not change; the prior defect rate did.

Bayes' Theorem with Three Sources

A retailer receives shipments from three warehouses.

Warehouse	Share of shipments	Late rate
W_1	0.50	0.04
W_2	0.30	0.06
W_3	0.20	0.10

A shipment is late. Which warehouse is most likely to have sent it?

Three Sources: Calculation

$$P(L) = 0.50(0.04) + 0.30(0.06) + 0.20(0.10) = 0.058$$

Warehouse	Joint probability	Probability given late
W_1	0.020	$0.020/0.058 \approx 0.345$
W_2	0.018	$0.018/0.058 \approx 0.310$
W_3	0.020	$0.020/0.058 \approx 0.345$

W_1 and W_3 are equally likely sources of a late shipment, despite their different late rates.

Exercise 5

Two suppliers deliver the same component.

Supplier	Share of deliveries	Defect rate
A	0.70	0.02
B	0.30	0.08

A delivered component is defective. Calculate the probability that it came from B.

Exercise 5: Solution

$$P(D) = 0.70(0.02) + 0.30(0.08) = 0.038$$

$$P(B | D) = \frac{P(B)P(D | B)}{P(D)} = \frac{0.30(0.08)}{0.038} = \frac{0.024}{0.038} \approx 0.632$$

Supplier B provides 30 percent of components but about 63.2 percent of defective components.

Exercise 6

A firm processes 1,000 orders.

- Four percent are fraudulent.
- A system flags 90 percent of fraudulent orders and 5 percent of legitimate orders.
- Reviewing a flagged order costs 15 MAD and prevents a 250 MAD loss if that order is fraudulent.
- Each fraudulent order not reviewed causes a 250 MAD loss.
- Compare the expected total cost of reviewing every flag with reviewing none.
- Should the firm review the flags?

Exercise 6: Solution

Expected counts for 1,000 orders under the given probabilities:

Actual status	Flagged	Not flagged	Total
Fraudulent	$40(0.90) = 36$	4	40
Legitimate	$960(0.05) = 48$	912	960
Total	84	916	1,000

$$P(\text{fraudulent} \mid \text{flag}) = \frac{36}{84} \approx 0.429$$

Expected total costs:

$$\text{Review flags: } 84(15) + 4(250) = 2,260 \text{ MAD}$$

$$\text{Review none: } 40(250) = 10,000 \text{ MAD}$$

Reviewing the flags gives expected savings of $10,000 - 2,260 = 7,740$ MAD under the stated costs.

What is next?

- Discrete Probability Distributions I
 - Random Variables
 - Developing Discrete Probability Distributions
 - Expected Value and Variance
 - Bivariate Distributions and Financial Portfolios
 - Reading(s): SBE Ch. 5.1 to 5.4

Exercise 7

An operations manager proposes investigating complaints only for late orders.

Delivery	Complaint	No complaint	Total
Late	35	65	100
On time	15	385	400
Total	50	450	500

Evaluate the proposal using the table. What should the manager investigate, and what cannot be concluded from these counts?

Exercise 7: Solution

$$P(L \mid C) = \frac{35}{50} = 0.70, \quad P(C \mid L) = \frac{35}{100} = 0.35, \quad P(C \mid L^c) = \frac{15}{400} = 0.0375$$

- Late orders account for 70 percent of complaints, but 15 of the 50 complaints came from on-time orders.
- Investigate both groups, with priority on late orders.
- These counts establish an association, not that lateness caused the complaints.

Exercise 8

An electronics manufacturer screens incoming components before assembly.

- Supplier records indicate that 1 percent are defective.
- The rapid test flags 95 percent of defective components and incorrectly flags 4 percent of acceptable components.
- A flagged component can undergo a manual check before it is used or discarded.

The operations manager proposes discarding every flagged component, claiming that a positive result makes a defect highly likely.

Evaluate this claim for a delivery of 10,000 components and recommend how to handle a positive result.

Exercise 8: Solution

Expected counts for 10,000 components under the given probabilities:

Actual status	Flagged	Not flagged	Total
Defective	95	5	100
Acceptable	396	9,504	9,900
Total	491	9,509	10,000

$$P(\text{defective} \mid \text{flag}) = \frac{95}{491} \approx 0.1935$$

- About 80.7 percent of flagged components are acceptable.
- Choosing between checking, using, and discarding requires the checking cost and the losses from discarding an acceptable component or using a defective one.

Exercise 9

A bank's audit team is reviewing six branches.

- Exactly two have elevated compliance risk, but the team does not know which ones.
- It randomly selects two branches for review.
- Before the detailed audit, a monitoring system checks the selected pair. It issues an alert with probability 0.80 if at least one selected branch is high risk, and 0.10 if neither is.

An alert occurs.

- The audit director wants to classify the pair as high risk without further verification, claiming that the alert proves one of the selected branches is high risk.
- Evaluate the claim and advise the audit team.

Exercise 9: Solution

Let H mean that the selected pair contains at least one high-risk branch. Then H^c means that neither selected branch is high risk.

- There are $\binom{6}{2} = 15$ possible pairs.
- Four branches are not high risk, so $\binom{4}{2} = 6$ pairs contain neither high-risk branch. The other $15 - 6 = 9$ pairs satisfy H .

Selected pair	Prior probability	Probability of alert	Joint probability with alert
H : at least one high-risk branch	$9/15 = 0.60$	0.80	$0.60(0.80) = 0.48$
H^c : neither high risk	$6/15 = 0.40$	0.10	$0.40(0.10) = 0.04$

$$P(\text{alert}) = 0.48 + 0.04 = 0.52, \quad P(H \mid \text{alert}) = \frac{0.48}{0.52} \approx 0.923$$

$$P(H^c \mid \text{alert}) = \frac{0.04}{0.52} \approx 0.077$$

- About 7.7 percent of alerts occur when neither branch is high risk.
- The alert does not prove the claim. Further verification depends on its cost and the consequences of a wrong classification.